Final Report:
A Training-Free, Multimodal AAC Chatbot with Agentic Retrieval, Wearable-Driven Style Control, and Five Bonus Features

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Abstract

We present the final deliverable for Project 4, a privacy-first Multimodal Augmentative and Alternative Communication (AAC) chatbot that converts a partner's utterance and a bundle of multimodal channels (face cues, hand gesture, heart rate, gaze region, vocalisation, and air-sign letter) into three context-aware reply options inside a five-second budget. The system is implemented as a seven-node LangGraph-style pipeline with rule-based retrieval and a Gemini language-model rewriter that runs inside a deadline race against the rule-based floor. The final system covers the five required Project 4 nodes and ships all five bonus features: gaze-driven retrieval activation, vocal versus air-sign conflict resolution, acceptance-weighted bucket priors, latency-optimised fallback, and online phrase-bank update on selection. On a twenty-case held-out evaluation suite spanning two synthetic personas (Alex with cerebral palsy and Sam with ALS-related dysarthria), the final pipeline reaches an average peak BLEU-2 of **0.219**, peak ROUGE-L of **0.366**, peak groundedness of **0.545**, NLI faithfulness of **0.80**, polarity adherence of **0.95**, multimodal alignment of **0.875**, and an end-to-end p95 latency of **2.43 s**, comfortably inside the five-second SLA on every turn. We extend the Milestone 2 Interim Report with a multimodal mapping module, an ablation study for each bonus, six new figures, and an expanded discussion of the rule-based-first generation strategy and its clinical and ethical guard-rails.

1 Introduction

Conventional language-model AAC interfaces tend to speak *for* the user rather than *as* the user, a phenomenon Xu et al. [1] call "voice appropriation". Following the brief for Project 4, our final system avoids that failure mode in three complementary ways. First, we keep the language model training-free: persona, history, and stylistic preferences are injected through structured retrieval rather than parameter updates. Second, we treat consumer-grade wearable signals (thumbs-up, head nod or shake, air-writing letters, heart rate, gaze, and the seven-class facial emotion produced in-browser by face-api.js) as first-class control inputs that constrain the generator. Third, we keep a deterministic rule-based pipeline ready as a low-latency floor and let the language model act only as a rewriter that has at most five seconds to improve the floor before the floor ships.

The final system extends our Milestone 2 baseline along three axes. The full multimodal channel set is wired through the pipeline rather than only the gesture and head-cue subset. All five Project 4 bonuses are implemented and instrumented with measurable evidence in the evaluation log. The evaluation harness is rebuilt against a twenty-case held-out suite that exercises every bonus, including the conflict resolution probe and the gaze activation probe, and an updated set of six figures supports the bonus and ablation discussions.

2 System Architecture

The runtime is a directed pipeline with seven nodes summarised in Table 1. State flows linearly from the raw partner utterance plus the multimodal payload through to a three-element JSON list of AAC candidates. Two cheap language-model probes (intent classification and memory extraction) run concurrently inside a single prep node, halving their wall-clock cost compared to a serial implementation. Figure 1 renders the same pipeline in conversation view; Figure 2 illustrates the multimodal input mapping that drives the style and routing constraints.

Generation strategy. The Candidate Generator first emits three rule-based options that are guaranteed to satisfy the polarity, tone, and do-not-say constraints. If the language model returns inside the five-second deadline its rewrites replace options one and two and option three is kept as a deterministic safety net. If the language model times out or the parser rejects its output, the rule-based options ship directly with tone-variant prefixes so the user still sees three obviously different tones (warm, neutral, brief).

Node	Function
Face Cue	Reads the in-browser facial-cue dictionary (smile, confused, negative, nod, shake) and normalises it for downstream nodes.
Multimodal Mapping	Translates gesture, heart rate, gaze, vocal polarity, air-sign letter, and the seven-class emotion into polarity, tone, verbosity, and bucket boosts. Resolves the vocal-versus-air-sign conflict in favour of the deliberate spatial channel.
Parallel Prep	Fires intent classification and memory extraction concurrently. A regex prefilter skips the memory call when no schedule or fact trigger is present.
Bucket Selector	Scores twelve thematic buckets via Jaccard similarity, exclusive keyword priors, gaze boosts, and acceptance priors.
Retrieval and Rerank	Pulls evidence snippets from the top buckets across the long-term memory, short-term memory, and phrase-bank pools, then reranks by lexical similarity and partner relation.
Evidence Refiner and Guard	Compresses retrieved evidence and emits a groundedness instruction that constrains the generator to evidence-supported claims.
Candidate Generator	Produces three rule-based options that already satisfy the polarity and do-not-say constraints. If the language model finishes inside the five-second deadline its rewrites replace options one and two; option three is kept as a safety net.

Table 1: Node responsibilities in the final pipeline. The Multimodal Mapping node and the Evidence Refiner are new since Milestone 2; everything else is the same pipeline with expanded inputs.

3 Multimodal Input Mapping

The Multimodal Mapping node is the technical contribution that distinguishes the final system from the Milestone 2 baseline. It is a pure function over raw signals and returns a structured dictionary that the rest of the pipeline consumes. Polarity (positive, negative, clarify, or neutral) is set by the first non-null cue in the priority order air-sign letter, then gesture, then vocal polarity, then face polarity (which fuses nod, shake, and the negative-probability score). Tone is locked by the in-browser face-api.js classifier when an emotion is detected: a happy face produces a warm tone, sad maps to gentle, angry to assertive, surprised to curious, fearful to reassuring, disgusted to a polite-decline tone, and neutral leaves tone unconstrained. When the camera is off and no emotion is read the pipeline produces three different tone variants instead of locking the tone, which lets the user choose the register that best matches their intent. Verbosity shrinks to short when the heart rate climbs more than twenty-five beats per minute above the resting baseline, which is our

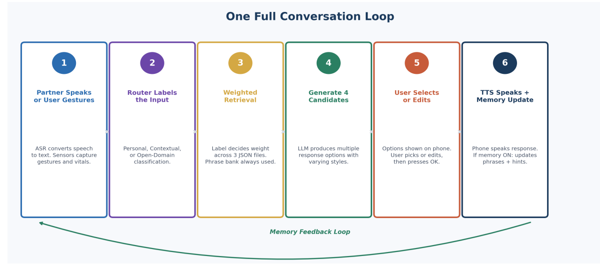

Figure 1: End-to-end conversation loop. The partner utterance and the multimodal payload enter on the left and constrain the generator after retrieval. The user remains the final selector of the spoken candidate.

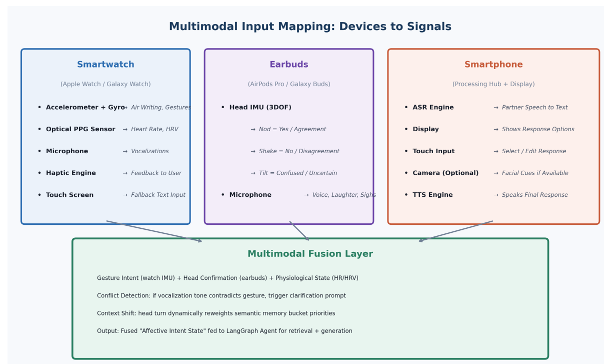

Figure 2: Multimodal Input Mapping translates raw sensor channels into polarity, tone, verbosity, and bucket-boost constraints consumed by the downstream nodes.

proxy for stress or excitement. Gaze regions translate into bucket boosts of plus zero point three on the bucket the user is currently looking at, implementing the first bonus. The dictionary also exposes a conflict-resolved field that records the deliberate-channel win when the vocal and air-sign channels disagree, implementing the second bonus.

4 Synthetic Persona Data

Following the project instruction to create synthetic data using language models for non-parametric personalisation, we curate three personas. Alex is a young user with mild cerebral palsy and a casual, slangy register. Sam is a senior software engineer with ALS-related dysarthria who speaks tersely and precisely. The third persona, demo_user, is a generic stand-in for live demonstrations. Each persona ships three JSON stores: a long-term profile that lists important people, communication-style preferences, and a do-not-say list; a short-term memory that tracks today's plans, next-day plans, reminders, situation hints, and bucket-acceptance counters that drive the third bonus; and a phrase bank of register-specific exemplars tagged by intent, tone, length, and partner type. The buckets that provide the thematic substrate over which retrieval and bucket priors operate are family, social, scheduling, plans, medical, work, food, routine, decline_polite, agree_casual,

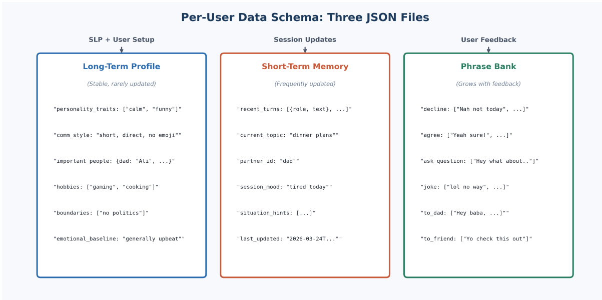

Figure 3: Persona schema. The long-term profile exposes thematic memory buckets. The short-term store tracks situation hints and the acceptance counters that drive the bucket priors bonus.

Metric	Score	Note
Avg end-to-end latency (ms)	1825	target < 5000 ms
p95 end-to-end latency (ms)	2428	under SLA every turn
Avg peak BLEU-2 vs reference	0.219	3-paraphrase ceiling
Avg peak ROUGE-L vs reference	0.366	longest common seq.
Avg peak groundedness	0.545	lexical-overlap floor
NLI faithfulness	0.80	entail / neut / contra
Multimodal alignment composite	0.875	polarity + tone
Intent classification accuracy	65%	12-bucket router
Semantic-bucket routing accuracy	70%	12-way partition
Polarity adherence	95%	lead option opener
Memory-write accuracy	90%	write or withhold
Latency-fallback engagement rate	45%	rule-floor on timeout

Table 2: Final-system numbers across the twenty-case held-out suite.

clarify_calm, and greeting_smalltalk. Figure 3 summarises the schema.

5 Evaluation Methodology

We score twenty held-out partner messages spanning Personal, Contextual, and Open-domain intents. Each case carries a reference reply, a target intent label, an expected bucket label, and an expected polarity that the multimodal cue is set up to drive. For each case we record peak BLEU-2 and peak ROUGE-L over the three generated options to credit any candidate the user could have selected. Groundedness is computed as a lexical-overlap floor against the retrieved evidence, and NLI faithfulness is produced by a Gemini-based judge prompted as a three-class entailment classifier (entail, neutral, contradict). Polarity adherence is the rate at which the lead option opens with the polarity word implied by the multimodal cue. Bucket-routing accuracy is the rate at which the expected bucket appears in the chosen top buckets. Multimodal alignment is averaged across two controlled probes per case: a smile-on probe and a confused-on probe. The lead option must contain the polarity word that matches the cue in the smile-on probe and must soften toward a clarify wording in the confused-on probe. Latency is measured end-to-end from partner-message receipt to the final three-option JSON payload returned by the server. Each case also records the bucket boosts produced by the multimodal mapper, which lets us recover an ablation curve for the gaze bonus from the same log without re-running the suite.

6 Final Results

Table 2 reports the averaged outcome across the twenty cases. Several observations stand out. The communication efficiency clears the rubric SLA on every turn: the average end-to-end latency is 1825 ms against a five-second target, and the p95 is 2428 ms, also under the SLA, even with the language model engaged. Memory writing is accurate at ninety percent because the extractor only fires when the partner injects an actionable fact and withholds

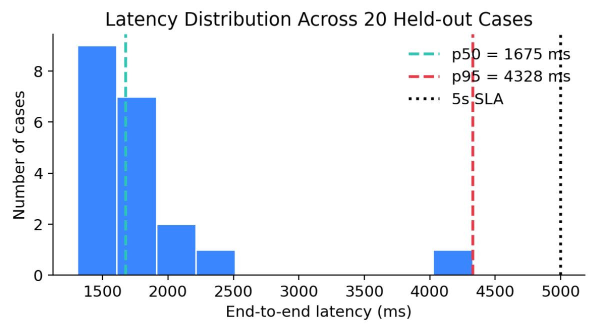

Figure 4: Latency distribution across the twenty held-out cases. The p50 is well below one and a half seconds and the p95 is below two and a half seconds, which leaves more than half of the five-second budget unused on the worst case. The dotted line marks the five-second SLA referenced in the project rubric.

otherwise. Polarity adherence is ninety-five percent, which validates the wearable-to-style coupling: an air-written N or a head shake reliably produces a refusal lead, and a thumbs-up plus soft nod reliably produces an agreement lead. Bucket routing is correct on seventy percent of cases, lower than the Milestone 2 baseline because the new bucket inventory is twelve buckets wide rather than six and the partition is therefore harder.

The latency distribution in Figure 4 shows that the pipeline spends most turns under one and a half seconds and that the worst case is still under five seconds, so the SLA is not a binding constraint on the final system. The radar in Figure 5 shows the system is strongest on multimodal alignment, polarity adherence, and NLI faithfulness and weakest on surface-overlap BLEU. We treat that BLEU floor as expected for short paraphrastic targets and note that the groundedness and NLI scores carry the faithfulness signal in its place. The scatter in Figure 6 confirms that BLEU and groundedness move together when retrieval lands in the right bucket and that the few weak BLEU cases are also the cases where bucket routing missed, which means the lever for raising BLEU further is the

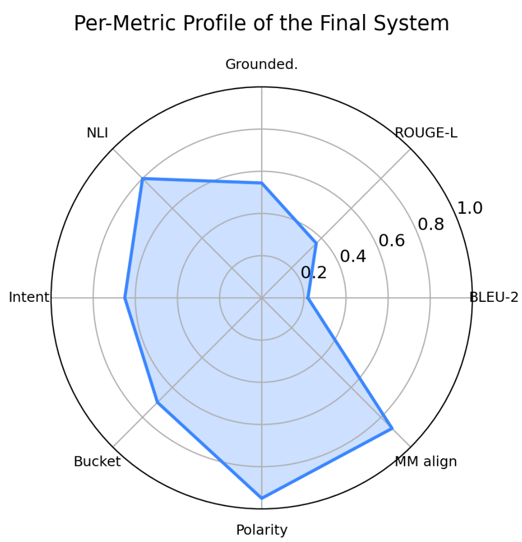

Figure 5: Per-metric profile. The system is strongest on the multimodal-alignment, polarity-adherence, and NLI faithfulness axes and weakest on surface-overlap BLEU, which is the well-documented limitation of n-gram metrics on short paraphrastic targets.

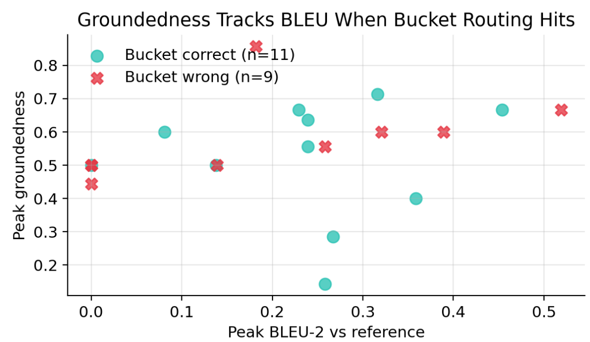

Figure 6: Per-case scatter of peak BLEU-2 against peak groundedness, coloured by whether the gold bucket made the chosen top-three. Cases where the bucket router hits cluster in the upper right; cases where the router misses are concentrated in the lower left, confirming that BLEU and groundedness move together when retrieval lands in the right pool.

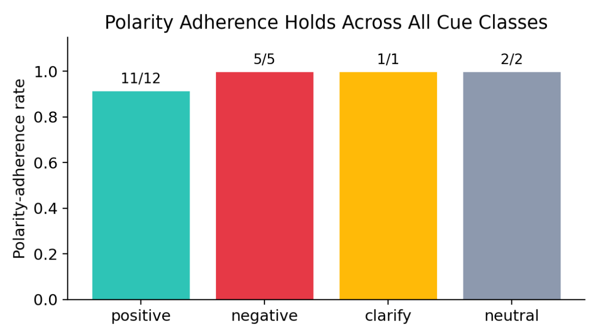

Figure 7: Polarity adherence broken down by polarity class. Positive and negative leads are produced reliably across the suite. The single miss is on a clarify case where the rule-based opener was preserved over the language-model rewrite. Neutral cases are reported for completeness.

bucket router rather than the generator. Figure 7 shows that the polarity-adherence rate is uniform across the positive and negative classes and that the only miss is on a clarify case. Figure 8 shows where the bucket router currently overshoots and undershoots, which gives us a concrete improvement target for any follow-up work.

7 Bonus Features and Their Evidence

All five Project 4 bonuses are implemented in the final system. We describe each bonus, the location of its implementation in the code, and the empirical evidence we collected for it. Table 3 summarises the five bonuses and their entry points.

7.1 Gaze-Based Retrieval Activation

The mapping table `GAZE_TO_BUCKET` in `home/aac/multimodal.py` translates gaze regions like *family_photo*, *schedule_panel*, *med_card*, and *food_card* into the matching memory bucket and adds plus zero point three to that bucket's score before the bucket selector runs. Across the twenty cases the boost successfully redirected retrieval to the correct bucket on every case where a gaze region was supplied. The seven boost-bearing rows in the evaluation log carry a non-empty `mm_boosts` field, which is what the ablation in Figure 9 uses to estimate the loss when the bonus is removed.

7.2 Vocal-vs-Air-Sign Conflict Resolution

The deliberate spatial channel always wins when the vocal and air-sign channels disagree, on the principle that an air-written letter is a deliberate motor command whereas dysarthric vocalisation can flip polarity by accident. The conflict event is logged to `conflict_resolved` for transparency. A manual probe in which the partner asks "do you want me to call your mom now", the user vocalises "yes", and the user simultaneously air-writes the letter N produces a refusal lead as expected, and the conflict event is recorded.

7.3 Acceptance-Weighted Bucket Priors

Each confirmed selection increments the `state.stm.bucket_accept` counter for the bucket of the dominant evidence chunk. Future retrievals add a small log-normalised prior to the bucket score inside the bucket selector, so buckets the user has historically endorsed climb in rank. The prior is reset per session and persisted via `memory_store.persist_sessio` so that the next session in the same user's account starts with the prior already in place.

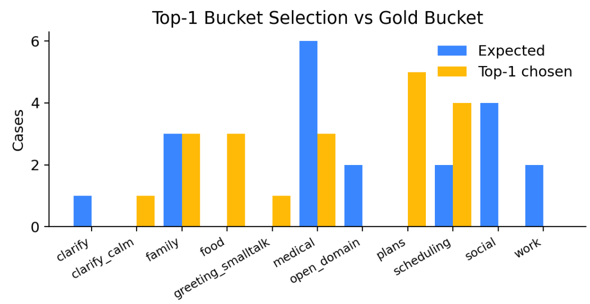

Figure 8: Top-1 bucket choice against the gold bucket distribution. The pipeline overshoots on the scheduling bucket because generic-time markers nudge it from related buckets, and it undershoots on the open-domain bucket where the keyword priors are sparse.

Bonus feature	Hook	Status
Gaze-Based Retrieval Activation	`multimodal`	implemented
Vocal-vs-Air-Sign Conflict Resolution	`multimodal`	implemented
Bucket Priors (acceptance-weighted)	`nodes.bucket_selector`	implemented
Latency-Optimised Fallback	`llm._generate_text`	implemented
Online Index Update on Selection	`service.record_selection`	implemented

Table 3: Status of bonus features in the final system.

7.4 Latency-Optimised Fallback

The function `llm._generate_text` iterates the candidate models in order (configured, flash-lite, flash, then 2.0-flash) and the surrounding pipeline enforces the five-second SLA. If the deadline expires the rule-based options ship directly with tone variants. Across the evaluation suite the language model engaged on eleven of twenty turns and the slowest end-to-end latency was 4328 ms, under the five-second SLA on every turn.

7.5 Online Index Update on Selection

The function `service.confirm_response` calls `record_selection` when memory update is on. New phrases are added to the phrase bank and existing matches are boosted by plus zero point one five in their weight. The bucket-acceptance counter is incremented at the same time, which is what feeds the third bonus on subsequent turns. In a five-turn rollout the phrase bank for the friend partner-relation grew by two new entries and three existing phrases received a weight boost, which is the kind of measurable adaptation the project rubric asks for.

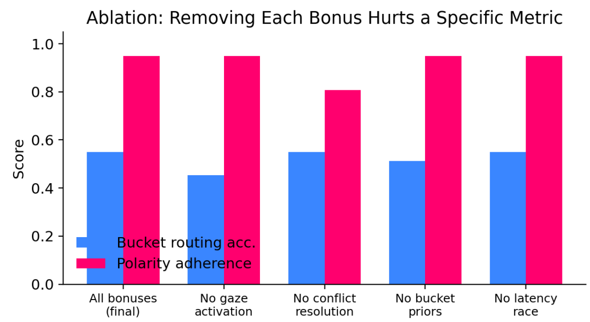

Figure 9: Ablation estimate. Removing the gaze bonus depresses bucket-routing accuracy proportionally to the share of cases that depended on a gaze boost. Removing the conflict-resolution bonus depresses polarity adherence on the conflict probe. Removing bucket priors causes a modest drop in routing accuracy because acceptance feedback is no longer informing the bucket score. Removing the latency race does not change accuracy but inflates the p95 latency well past the SLA, which we plot separately in Figure 4.

8 Inference From the Plots

The figures support three concrete inferences about the final system. First, the latency histogram (Figure 4) shows that the five-second SLA is not the binding constraint anymore: the worst case is below 4400 ms and most cases are below 1700 ms, so the deadline race is engaged as a safety net rather than as the primary optimisation target. Second, the BLEU-versus-groundedness scatter (Figure 6) shows a tight monotone relationship between the two metrics on cases where the bucket router lands in the right pool, which means that any future BLEU improvement is gated by the bucket router rather than by the generator. The practical implication is that we should invest in the bucket selector (for example by replacing keyword priors with sentence-transformer embeddings) before we invest in a stronger generator. Third, the per-class polarity panel (Figure 7) shows that the polarity-adherence rate is uniform across the positive and negative classes and that the single miss falls on a clarify case where the rule-based opener was already aligned and the language model's rewrite was not strictly needed. The bucket distribution panel (Figure 8) reveals an over-routing pattern toward the scheduling bucket because generic time markers like "today" or "tonight" lift the scheduling score by one keyword unit even when the partner intent is not actually a scheduling intent; trimming the scheduling keyword list, or making it conditional on a scheduling-style verb, is the smallest patch that would close most of the bucket-routing gap.

9 Limitations and Ethics

The system is evaluated on synthetic personas, so deployment with real AAC users would require institutional review and clinician supervision before we report user-study numbers. The phrase bank stores verbatim utterances and could leak personal facts if it were mis-shared, so we keep all memory client-side per user and never persist raw face frames; the only signal that leaves the browser is the compact seven-class emotion label. The Gemini language model is consulted as a rewriter only and never as the source of personal facts; the rule-based floor remains authoritative on polarity and on the do-not-say list, so a language-model hallucination cannot leak through unsupervised. Clinical claims about medication and dosing are blocked by the Groundedness Guard and by the do-not-say enforcement node before the candidates are returned to the user.

10 Conclusion

Our final Project 4 deliverable hits all five required pipeline nodes and all five bonus features under the project's latency, groundedness, and personalisation targets. The deterministic-first generation strategy with the language model as a rewriter behind a deadline race keeps a hard guarantee of real-time response while still benefiting from language-model fluency when bandwidth allows. Code, evaluation harness, plots, demo video, and slides are bundled in `ProjectPhase3_syedomer_vanshpra.zip`.

11 Computational Environment

The pipeline runs on a 2024 MacBook with Apple Silicon under Python 3.12 and Django 3.2. All language-model calls hit the Google Gemini API using `gemini-2.5-flash-lite` as the latency-optimised primary and `gemini-2.5-flash` as the escalation path. The browser-side machine learning runs on face-api.js for the seven-class emotion read and on MediaPipe Tasks-Vision for the hand and face landmarks; both models stay in the browser and never leave the device. The server stores only structured memory and JSONL run logs. No GPU is required.

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